DETAILED PROGRAM

MONDAY 21:
Introduction to Legged Robots (Michele Focchi) (5h)

A1. Introduction to floating based robots
A2. Modeling of legged robots: dynamics of floating base robots (excess of coordinates/minimal coordinates modeling)
A3. Building a simulator: contact Models (rigid contact, compliant contacts), modeling of impact dynamics. Constrained dynamics obtained by contact null-space projection.
A4. From floating base dynamics toward Newton-Euler equations
A5. Basic Concepts in Legged Locomotion
A6. Quasi-static Trunk Controller
A7.Reduced models: Linear Inverted pendulum (LIP), Centroidal Dynamics, Single Rigid Body model, point mass model, inverted pendulum, spring-loaded inverted pendulum (SLIP)
A8. Balance metrics: Capture point
A9. Locomotion Frameworks

TUESDAY 22:

Primer on optimization/optimal control (morning)(Marco Frego) (3h)

B0. introduction to numerical optimization: LP, QP, SOCP (convex), non linear
B1. Taxonomy explain direct/indirect method. Overview on Optimal control Problems transcription: Single shooting / Multiple Shooting/ Direct collocation (piece-wise knot, ZoH) for CoM trajectory optimization, examples single shooting, direct collocation.
B2. Automatic differentiation/finite differences/mention analytical derivatives of Carpentier for rigid body dynamics
B3. Types of Solvers: SQP(example), Interior Point, Active set, introduce sparsity concept
B4. Discretization of dynamics (forward Euler, exact discretization with matrix exponentials (for linear systems), RK4

Advanced robot control (afternoon) (Andrea Del Prete)(3h)

C1. Constraint inverse dynamics for floating base robots.
C2. Whole-body control: Quasi-static approach (pseudo-inverses) to control the Base or the Com
C3. QP-based controller (with inequality constraints): joint space/task space inverse dynamics (Torque limits, acceleration limits, velocity limits, position limits)
C4. QP-based controller in contact: friction constraints

WEDNESDAY 23:

Trajectory Optimization (morning) (Enrico Mingo) (3h)

E1. Reactive vs. predictive control: differential inverse kinematics as a QP
E2. Whole-body control and contact modeling
E3. Linear Inverted Pendulum Model (LIPM) and ZMP
E4. Linear optimal control with LIPM (time invariant), sparsity and Riccati recursion
E5. Trajectory optimization vs. MPC: LIPM MPC with automatic footstep placement

E6. Linear optimal control and MPC with Single Rigid Body Dynamics (time varying)
E7. Non-linear trajectory optimization with full models via SQP
E8. Non-linear MPC via SQP, contact-implicit TO and robust MPC

Sampling-based Control (afternoon) (Giulio Turrisi)(3h)

F1. Sample-based approaches for ocp solution
F2. Random Sampling
F3. Splines, Warm Start, CPU-GPU
F4. CEM
F5. MPPI
F6. Application on legged robots

THURSDAY 24:

State Estimation for legged robot (morning) (Marco Camurri) (3h)
H1. Introduction to state estimation and SLAM
H2. Recursive Bayes filter
H3. Kalman filter and Extended Kalman filter
H4. Lie groups and Invariant EKF
H5. Smoothing with factor graphs and robust least squares
H6. Inertial-legged state estimation: leg odometry, contact estimation and preintegration

Reinforcement Learning (Afternoon) (Giulio Turrisi) (3h)
G1. Introduction to Reinforcement Learning:
G2. Value Iteration
G3. Q-Iteration
G4. Q-Learning
G5. Deep Q-Learning
G6. Policy Gradient
G7. PPO

Social Dinner + delivery of the diplomas

FRIDAY 25:

LABS (Friday Morning)(3h)
Locomotion lab in **simulation**
Reinforcement Learning (Friday afternoon) (Giulio Turrisi) (3h)
H1. Simulators in RL
H2. Rewards
H3. Observations
H4. IMU vs Linear velocity vs Concurrent SE
H4. Sim2Real
H5. Asymmetric PPO

SATURDAY 26:
Optional tutored practical session with **Real Robots** Aliengo/go2

SUNDAY 27:
Trekking in Dolomites